A) What Wisconsin data implies (to anchor the salt side)

- **Salt used (WI 2023-24): 7.3 tons per lane-mile for the season.** Wisconsin Department of Transportation+1
- **Typical granular salt rate: ~172 lb per lane-mile per application** (NYSDOT public figure; close to other DOTs). NYSDOT+1

Implied number of passes:
7.3 tons × 2000 lb / 172 lb ≈ **~85 applications per season**.
If WI had ~**~21 winter events**, that's **≈ 4 passes per storm** on average. *(This is a math inference using WI + NY inputs; that's why investors see 2–4 passes per storm as typical.)*
Wisconsin Department of Transportation+1

Salt works best above ~15–20°F. Below that, its "practical" performance falls off.
Cargill+2Nova Region+2

B) One-storm, danger-zone math (not miles)

Let's define a **danger-zone segment** as **0.25 lane-mile** (a reasonable overpass length).

Salt (per pass, per 0.25 LM)

- **Materials:** 172 lb/LM/app × \$0.0456/lb (WI 2023-24 salt price) = **\$7.85/LM** → **\$1.96** for 0.25 LM.
- **Labor+truck (midpoint):** ≈ **\$6.00/LM** → **\$1.50** for 0.25 LM.
- **Per pass total: \$3.46** per 0.25 LM.

- **Per storm typical:**
  - **2 passes: $6.92**
  - **3 passes: $10.38**
  - **4 passes (WI-implied avg): $13.84**

(Salt price from WI 2023-24; rate from NYSDOT 2024-25; "passes per storm" is the WI/NY derivation above.) Wisconsin Department of Transportation+1

Pre-De-Icer (per 0.25 LM, one preventive pass)

- Your **bulk chemical cost:** $**9.65/gal** (your $12.86 less 25%).
- Labor+truck same midpoint: **$1.50** for 0.25 LM.
- **Material cost depends on rate** (gal/LM). The segment cost is:
  Segment cost = $1.50 + 0.25 × ($9.65 × rate_gal/LM)

PDI rate (gal/LM)	Segment materials	Total per segment (1 pass)
2.0	$4.83	**$6.33**
3.0	$7.24	**$8.74**
4.0	$9.65	**$11.15**

Takeaway:

- If salt would be **2 passes** on that bridge, **PDI beats salt at ≲ 2 gal/LM** ($6.33 vs $6.92).
- If salt is **3 passes**, PDI is competitive up to **≈ 4 gal/LM** ($11.15 vs $10.38).
- If salt is **~4 passes** (WI-implied average), PDI wins **even at ~4 gal/LM** ($11.15 vs $13.84).

- **And** PDI still works **below 27°F** where salt under-performs. Cargill+1

This is exactly why we recommend **PDI only for danger zones** (or a short main-street stretch), **not every mile** of open highway.

Positioning sentence you can paste:

> "Pre-De-Icer is a **premium, preventive** treatment for **danger zones**—bridges and overpasses where crashes spike and salt often needs **3–4 reactive passes**. Using our bulk cost ($9.65/gal) and conservative labor, **one PDI pass** is cost-competitive when salt requires multiple passes (2–4), and it maintains performance at temperatures where salt is less effective. We are **not** proposing PDI for miles of rural highway; we're proposing **targeted segments** and the **consumer windshield** product."

D) "Other pre-icers" claim (why PDI is different)

- **Commodity "de-icers" for windshields** are typically **alcohol-heavy, reactive** sprays (e.g., methanol/ethanol based; see Prestone SDS). They **melt after** the bond forms and can be harsh on materials. PDI is a **preventive coating** with viscosity/adhesion that **stops bonding** and **coats blades** so they don't freeze mid-drive. Mighty Auto Parts+1
- **Brines** (liquid NaCl) are cheap but also **salt-temperature-limited** (15–20°F practical). Cargill+1

Short version: "We aren't selling PDI by the mile. We're using it where it **matters most**. On a **0.25 lane-mile bridge**, salt usually needs **2–4 passes** (Wisconsin + NY rates imply ~4 on average), which runs **$6.92–$13.84** per storm just in this segment. **One PDI pass** costs **$6.33–$11.15** depending on application rate (2–4 gal/LM) and still works below 27°F. That's why PDI is the **right tool for danger zones**—and why the **windshield** product is our **fastest-to-market** consumer win."